jitters, and so on are directly incorporated. In fact, in Chapter 12 we have seen this procedure in action: The loop processes (see Figure 12.6) incorporate quantum concepts—in this case, the momentary quantum-mechanical creation of virtual string pairs—with the number of loops determining the precision with which quantum-mechanical effects are accounted for.

The strategy of beginning with a theoretical description that is classical and then subsequently including the features of quantum mechanics has been extremely fruitful for many years. It underlies, for example, the standard model of particle physics. But it is possible, and there is growing evidence that it is likely, that this method is too conservative for dealing with theories that are as far-reaching as string theory and M-theory. The reason is that once we realize that the universe is governed by quantum-mechanical principles, our theories really should be quantum mechanical from the start. We have successfully gotten away with starting from a classical perspective until now because we have not been probing the universe at a deep enough level for this coarse approach to mislead us. But with the depth of string/M-theory, we may well have come to the end of the line for this battle-tested strategy.

We can find specific evidence for this by reconsidering some of the insights emerging from the second superstring revolution (as summarized, for example, by Figure 12.11). As we discussed in Chapter 12, the dualities underlying the unity of the five string theories show us that physical processes that occur in any one string formulation can be reinterpreted in the dual language of any of the others. This rephrasing will at first appear to have little to do with the original description, but, in fact, this is simply the power of duality at work: Through duality, one physical process can be described in a number of vastly different ways. These results are both subtle and remarkable, but we have not yet mentioned what may well be their most important feature.

The duality translations often take a process, described in one of the five string theories, that is strongly dependent on quantum mechanics (for example, a process involving string interactions that would not happen if the world were governed by classical, as opposed to quantum, physics) and reformulate it as a process that is weakly